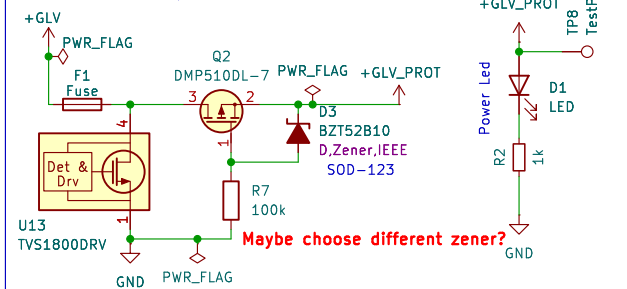
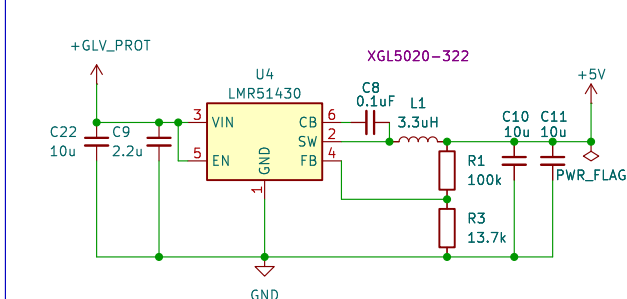


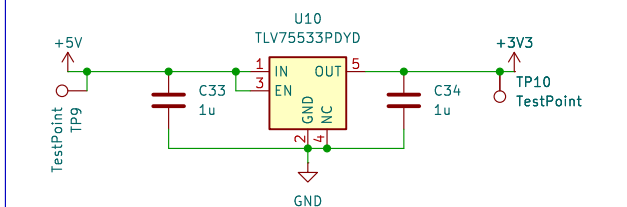
Reverse Polarity Protection



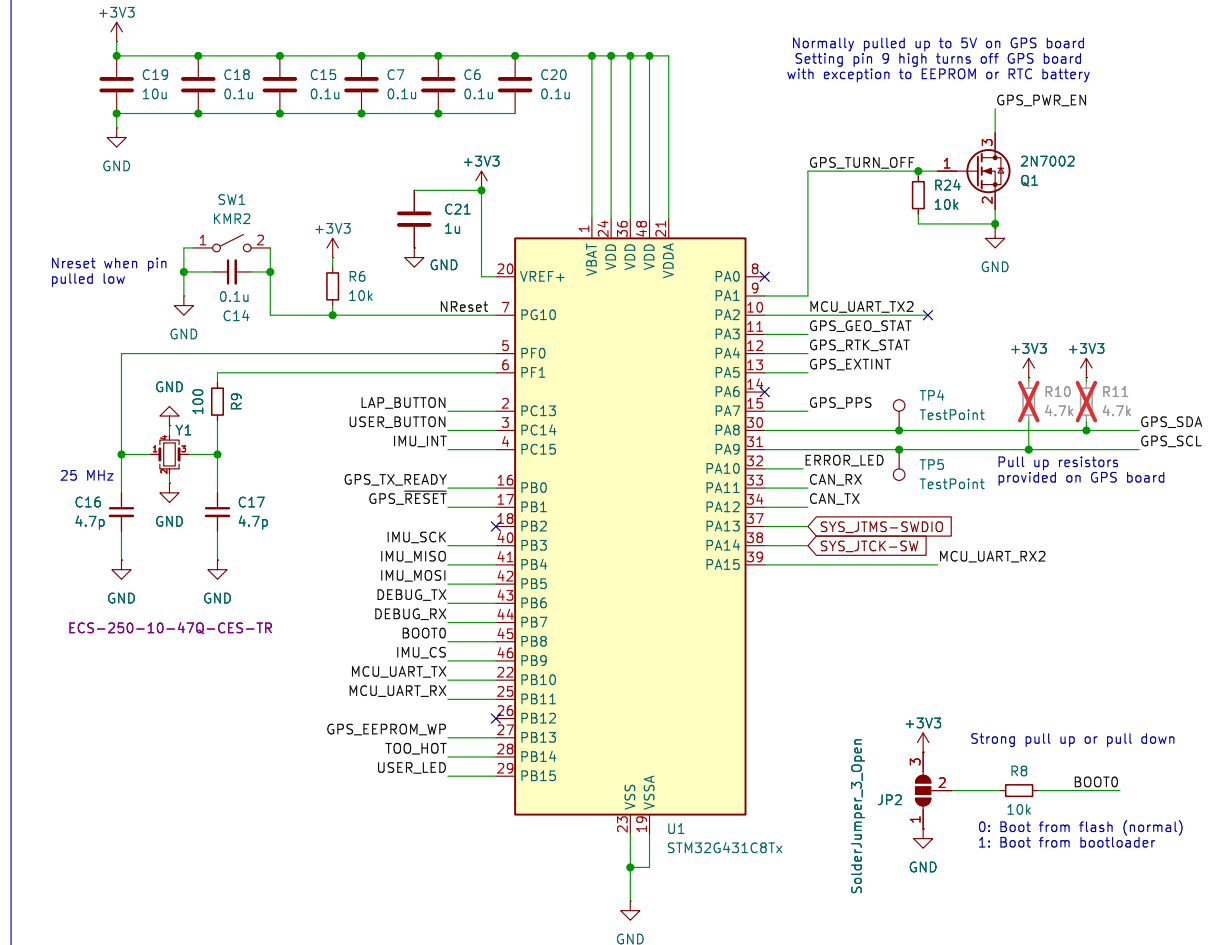
GLV - 5V Buck Converter



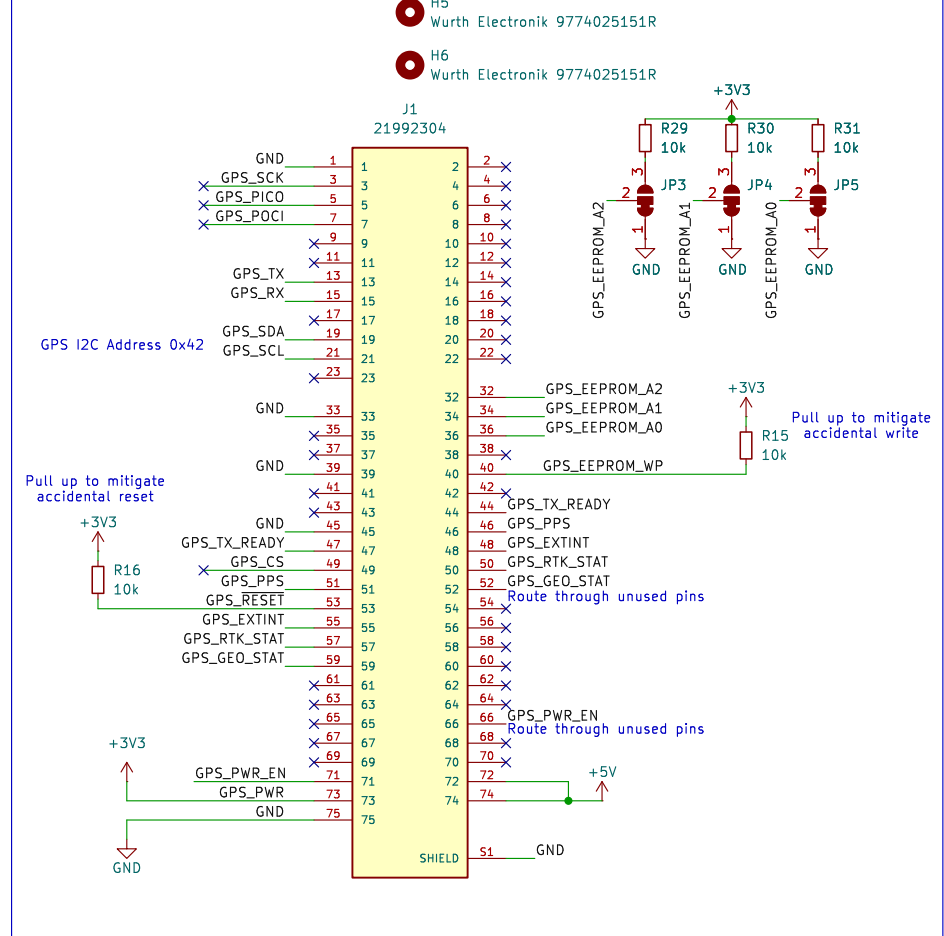
5V - 3.3V LDO



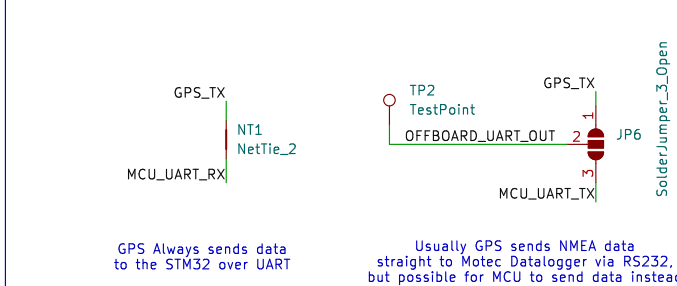
STM32 MCU



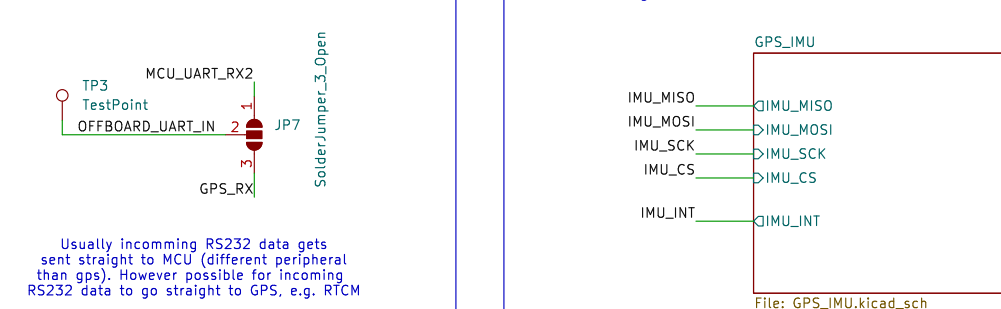
Micromod Connector



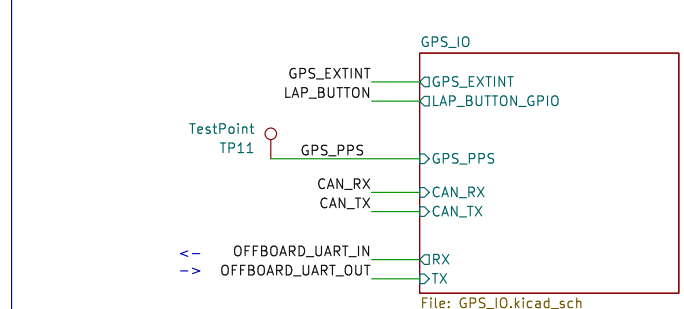
UART Direction Selection



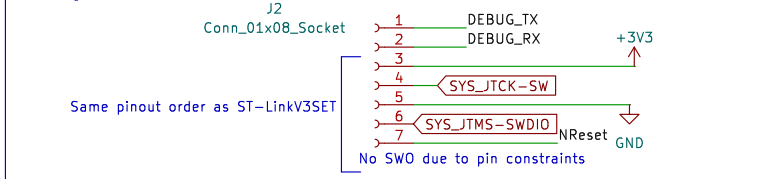
IMU and Magnetometer



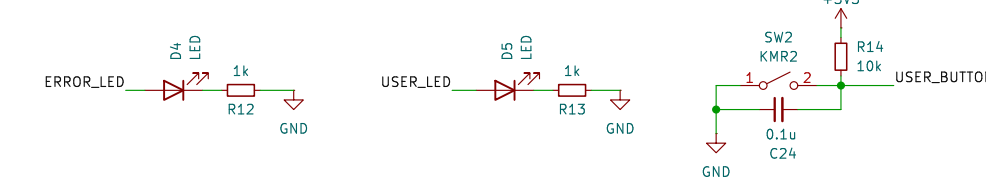
Inputs and Outputs



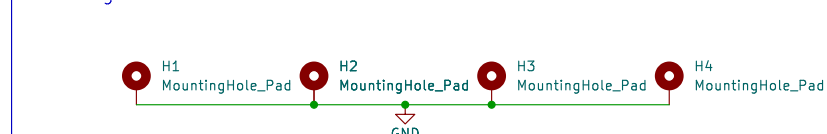
Debug



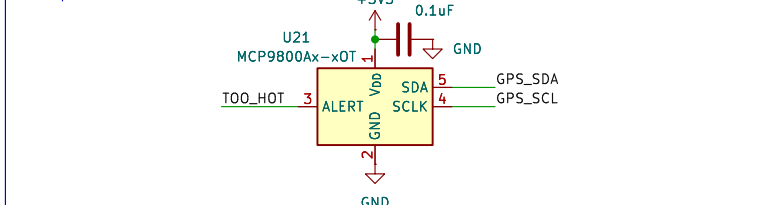
LEDs & Button



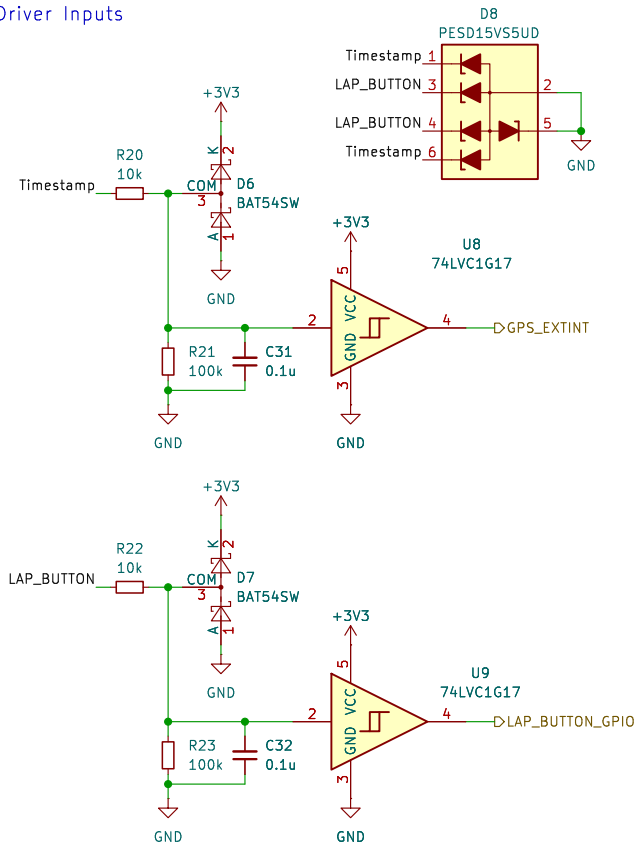
Mounting Holes



Temperature Sensor

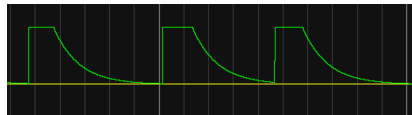


Driver Inputs



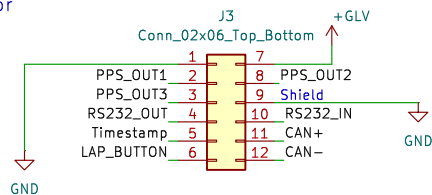
Signal conditioning of inputs.

Timestamp and Lap Reset inputs are capable of 24V
Current limiting resistor and BAT54SW bounds signal to 3.3V
RC filter to debounce. Tau=10ms
Schmitt trigger for precise turn on thresholds

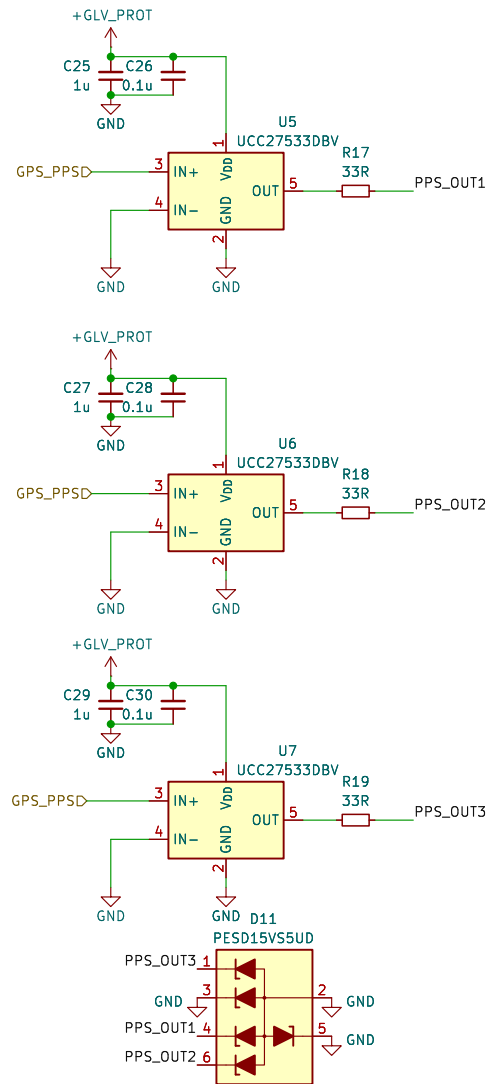


<https://is.gd/Ufd4W0>

Connector



PPS Signal Drivers

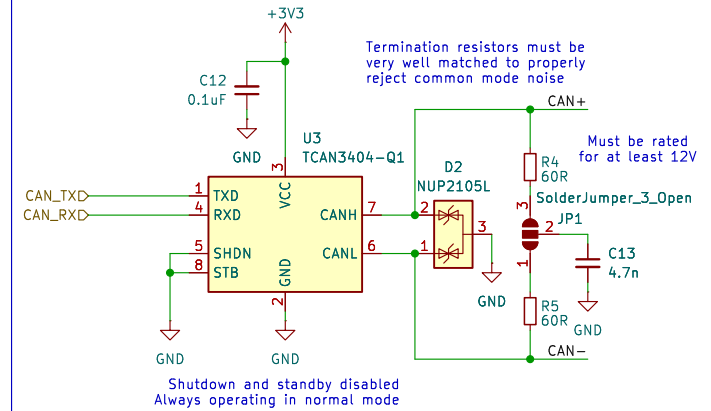


GPS sends a PPS signal every second precise to the microsecond.

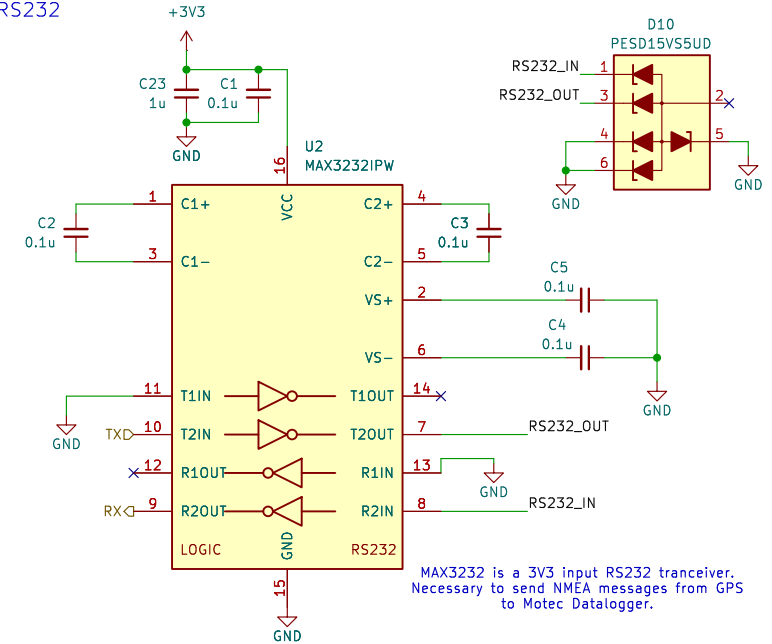
In order to distribute this signal to devices in the car, gate drivers are used, rather they are used as line drivers.

The gate drivers are capable of fast switching of GLV, essentially level shifting the PPS signal, making it more noise robust and capable of driving many devices.

CAN Trceiver



RS232



Vehicle: STAG 12

Drawn By: Adam Eastman

Checked By: Annabel Owen, Ray Wang

CAD Part:

SUFST

Sheet: GPS_IO

File: GPS_IO.kicad_sch

Title: GPS Mainboard

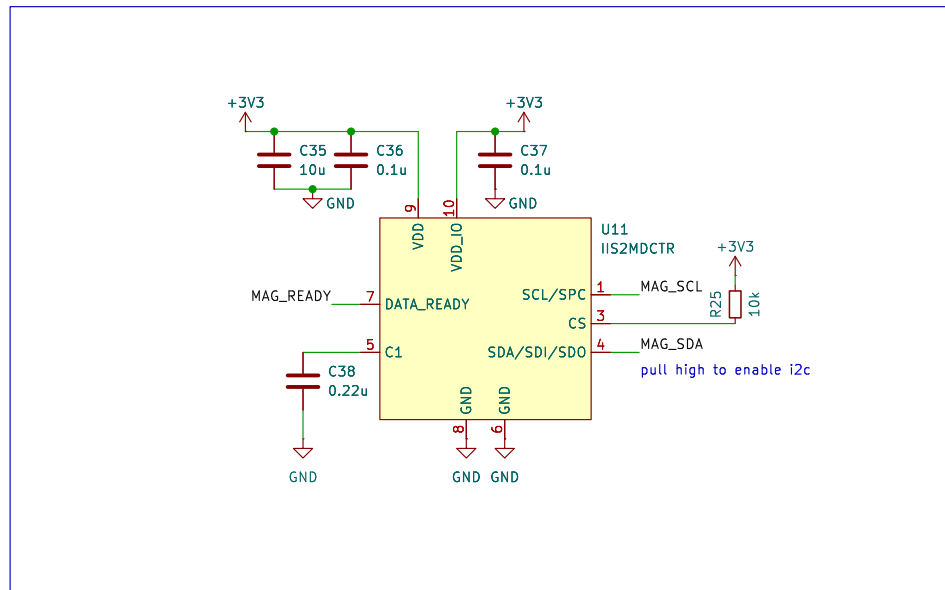
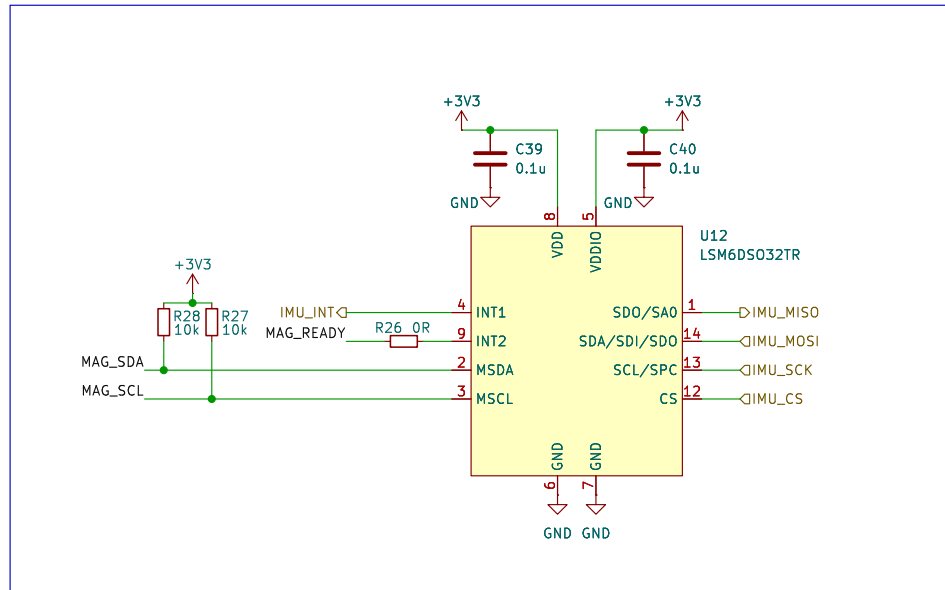
Size: A4 Date: 2026-04-01

KiCad E.D.A. 9.0.9.1

Rev: v1.0.0

Id: 2/3





Vehicle: STAG 12
 Drawn By: Adam Eastman
 Checked By: Annabel Owen, Ray Wang
 CAD Part:

SUFST

Sheet: GPS_IMU
 File: GPS_IMU.kicad_sch

Title: GPS Mainboard

Size: A4 Date: 2026-04-03

KiCad E.D.A. 9.0.9.1

Rev: v1.0.0

Id: 3/3

